GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	3.0
GEGN588	ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN590	GIS-BASED REAL WORLD LEARNING PROJECT I - FUNDAMENTALS	3.0
GEGN592	GIS-BASED REAL WORLD LEARNING PROJECT II - ADVANCED APPLICATIONS	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
GPGN545	INTRODUCTION TO DISTRIBUTED FIBER-OPTIC SENSING AND ITS APPLICATIONS	3.0
GPGN558	SEISMIC DATA INTERPRETATION AND QUANTITATIVE ANALYSIS	3.0
MNGN502	GEOSPATIAL BIG DATA ANALYTICS	3.0

Certificate Programs in Data Science

Program Requirements

There are six Certificates in Data Science. Applicants for each are required to have an undergraduate degree to be admitted into the Certificate programs. Course prerequisites, if any, are noted for each Certificate program.

Students working toward one of the Data Science Certificates are required to successfully complete 12 credits, as detailed below for each Certificate. The courses taken for the Certificates can be used towards a Master's or PhD degree at Mines, however courses used for one Data Science Certificate cannot also be counted toward another Data Science Certificate.

Graduate Certificate in Data Science - Statistical Learning (12 credits)

The Data Science - Statistical Learning Graduate Certificate is an online or residential program focusing on statistical methods for interpreting complex data sets and quantifying the uncertainty in a data analysis. The Certificate also includes gaining new skills in computer science but is grounded in statistical models for data, also termed statistical learning, rather than algorithmic approaches. Students will develop an essential skill set in statistical methods most commonly used in data science along with the understanding of the methods' strengths and weaknesses. Moreover, the coursework will cover a broad range of applications making it relevant for varied scientific and engineering domains.

Applicants must have completed the following courses, or their equivalents, with a B- or better: CSCI261 and CSCI262 Data Structures, MATH332 Linear Algebra and MATH334 Introduction to Probability.

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI530	STATISTICAL METHODS I	3.0
DSCI560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I	3.0
DSCI561	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II	3.0

Graduate Certificate in Data Science - Earth Resources (12 credits)

The Graduate Certificate in Data Science - Earth Resources is an online program building on the foundational concepts in data science as it pertains to managing surface and subsurface Earth resources and on specific applications (use cases) from the petroleum and minerals industries as well as water resource monitoring and remote sensing of Earth change. The Certificate includes one core introductory Data Science course, two courses specific to Earth resources and one elective.

DSCI503	ADVANCED DATA SCIENCE	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
ELECTIVE	(1) ELECTIVE FROM LIST BELOW	3.0

Graduate Certificate in Data Science - Earth Resources Electives (select ONE (1) from the list below):

Geospatial Focus:		
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN579	PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS	3.0
Petroleum Focus:		
GPGN519	ADVANCED FORMATION EVALUATION	3.0
GPGN547	PHYSICS, MECHANICS, AND PETROPHYSICS OF ROCKS	3.0
GPGN558	SEISMIC DATA INTERPRETATION AND QUANTITATIVE ANALYSIS	3.0
GPGN651	ADVANCED SEISMOLOGY	3.0
PEGN522	ADVANCED WELL STIMULATION	3.0
PEGN551	PETROLEUM DATA ANALYTICS - FUNDAMENTALS	3.0
Mining Focus:		
MNGN548	INFORMATION TECHNOLOGIES FOR MINING SYSTEMS	3.0
Hydrology Focus:		
CEEN581	WATERSHED SYSTEMS MODELING	3.0
Additional Options:		
DSCI/MATH530	STATISTICAL METHODS I	3.0
EBGN525	BUSINESS ANALYTICS	3.0

Graduate Certificate in Petroleum Data Analytics (12 credits)

The Graduate Certificate in Petroleum Data Analytics is an online program building on the foundational concepts in statistics and focusing on the data foundation of the oil and gas industry, the challenges of Big Data to oilfield operations and on specific applications (use cases) for petroleum analytics. The Certificate includes two core introductory Data Science courses and two course specific to petroleum engineering.

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI530	STATISTICAL METHODS I	3.0

PEGN551	PETROLEUM DATA ANALYTICS - FUNDAMENTALS	3.0
PEGN552	PETROLEUM DATA ANALYTICS - APPLICATIONS	3.0

Graduate Certificate in Business Analytics

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The certificate is an online or residential program. The
 requirements are to complete the core course and two elective
 courses:
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Core Course

EBGN525	BUSINESS ANALYTICS	3.0

Elective Courses

EBGN559	SUPPLY CHAIN ANALYTICS	3.0
EBGN560	DECISION ANALYTICS	3.0
EBGN571	MARKETING ANALYTICS	3.0

Course substitutions may be approved on a case-by-case basis by the certificate director. Completing the certificate will also position students to apply to either the master of science in engineering and technology management degree or the master of science in data science degree, as the certificate courses can be applied to either degree.

Graduate Certificate in Data Science - Foundations (12 credits)

The Data Science - Foundations Graduate Certificate is an online or residential program focusing on the foundational concepts in statistics and computer science that support the explosion of new methods for interpreting data in its many forms. The Certificate balances an introduction to data science with teaching basic skills in applying methods in statistics and machine learning to analyze data. Students will gain a perspective on the kinds of problems that can be solved by data intensive methods and will also acquire new analysis skills outside of the certificate. Moreover, the coursework will cover a broad range of applications, making it relevant for varied scientific and engineering domains.

Applicants must have completed the following courses, or their equivalents, with a B- or better: CSCI261 and CSCI262 Data Structures, MATH332 Linear Algebra and MATH334 Introduction to Probability.

DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
MATH560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I	

Graduate Certificate in Data Science - Computer Science (12 credits)

The Data Science - Computer Science Graduate Certificate is an online or residential program focusing on data science concepts within computer science (e.g., computational techniques and machine learning) plus prerequisite knowledge (e.g., probability and regression). The aim of this certificate is to help students develop an essential skill set in data analytics, including (1) deriving predictive insights by applying advanced statistics, modeling, and programming skills, (2) acquiring in-depth knowledge of machine learning and computational techniques, and (3) unearthing important questions and intelligence for a range of industries, from product design to finance.

Applicants must have completed the following courses, or their equivalents, with a B- or better: CSCI261 and CSCI262 Data Structures, MATH213 Calculus III and MATH332 Linear Algebra. DSCI530 Statistical Methods I, will serve as the MATH201 Probability and Statistics prerequisite for the two machine learning courses of the certificate (DSCI570 Introduction to Machine Learning and DSCI575 Machine Learning).

DSCI503	ADVANCED DATA SCIENCE	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
DSCI575	ADVANCED MACHINE LEARNING	3.0

Courses

DSCI503. ADVANCED DATA SCIENCE. 3.0 Semester Hrs.
(I, II) This course will teach students the core skills needed for gathering, cleaning, organizing, analyzing, interpreting, and visualizing data. Students will use the python programming language and related toolkits for data manipulation and the use and application of statistical and machine learning for data analysis. The course will be primarily focused on applications, with an emphasis on working with real (non-synthetic) datasets. Students will propose and design a semester project using a dataset from their domain of interest, leveraging the concepts and skills acquired from this course (e.g., data analysis, ethical considerations, evaluation and synthesis of results, storytelling and visualization). Prerequisite: CSCI200 with a grade of C- or higher or CSCI262 with a grade of C- or higher, MATH201 or MATH334 OR Graduate level standing and at least CSCI128 or equivalent.
Course Learning Outcomes

- Conduct data acquisition using a varied set of techniques structured and unstructured datasets; including raw data files, SQL databases, online repositories, and programmatically through web scraping and APIs.
- Apply preprocessing strategies to complex and dynamic datasets using industry-standard toolkits and machine learning algorithms to extract features, reduce dimensionality, remove errors, inconsistencies, and missing values.
- Differentiate between machine learning approaches such as classification, regression, clustering, and neural networks for predictive analytics and pattern recognition.
- Evaluate the predictive power of the different statistical and machine learning methods to solve real-world data science problems.
- Develop storytelling and visualization techniques to effectively communicate (exploratory) or persuade (explanatory) findings to a specific audience.
- Critically asses ethical considerations and challenges related to data collection and analysis.
- Construct a comprehensive data science project from inception to presentation, integrating the various techniques and tools learned throughout the course.

DSCI530. STATISTICAL METHODS I. 3.0 Semester Hrs.
Introduction to probability, random variables, and discrete and continuous probability models. Elementary simulation, data summarization and analysis using the R Data Analysis Environment. Confidence intervals and hypothesis testing for means and variances. Chi square tests.